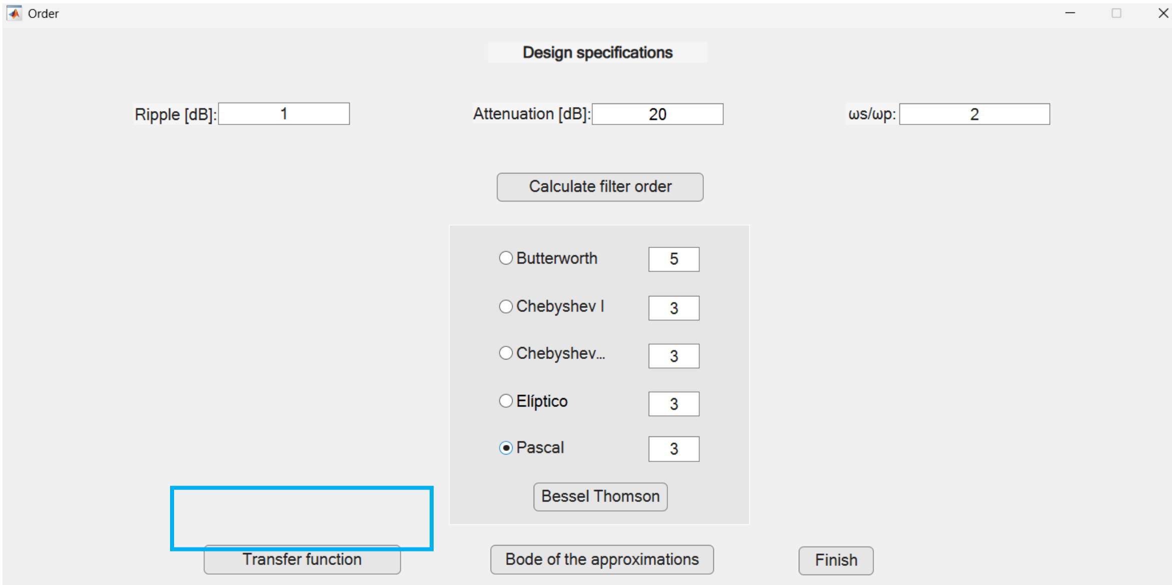
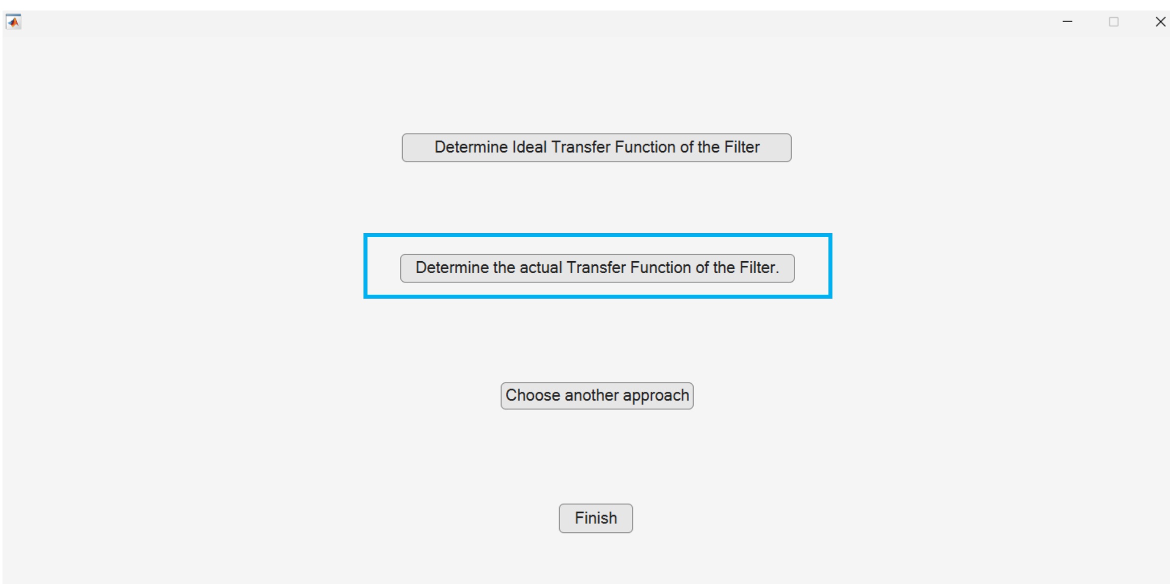
Design of Pascal band-pass filter for EMG signals

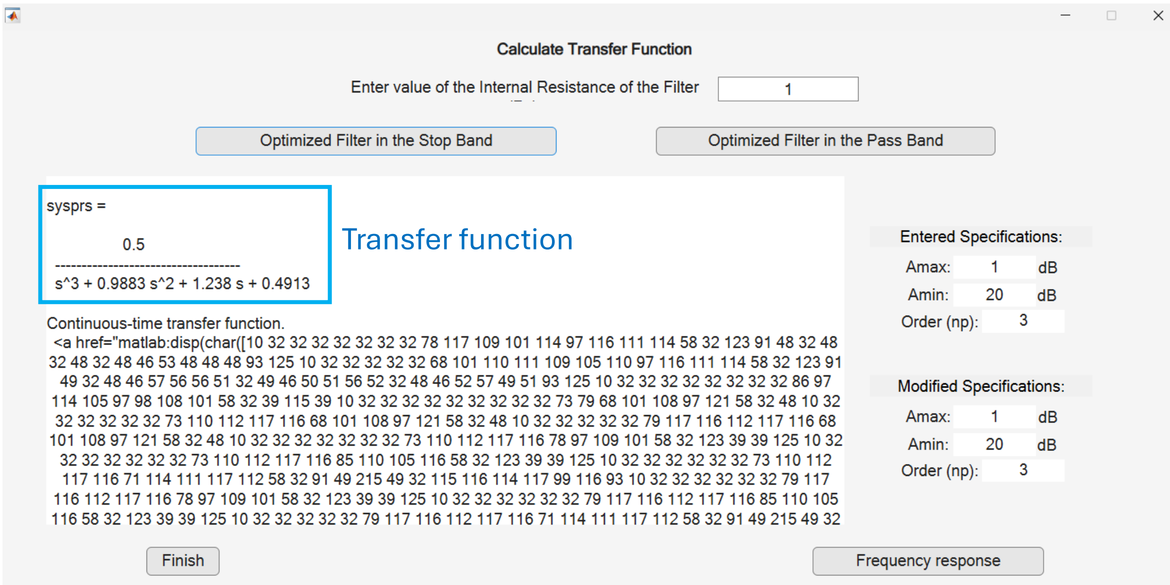
The interface is opened and the design specifications are entered.



Pascal is selected and the transfer function of the real case is calculated.



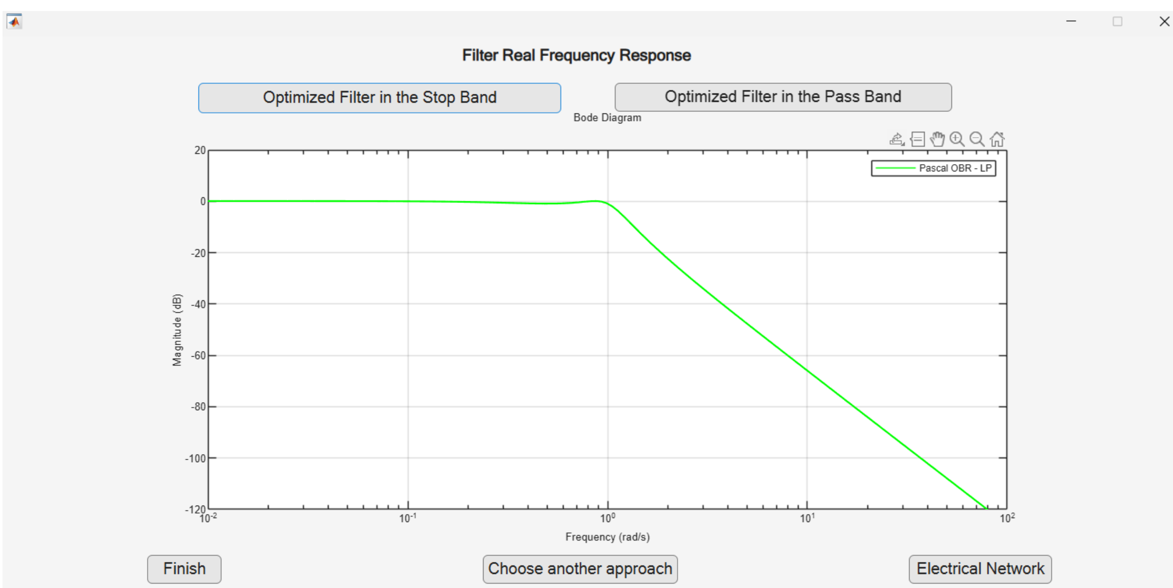
The option optimized filter in the stop band is selected.



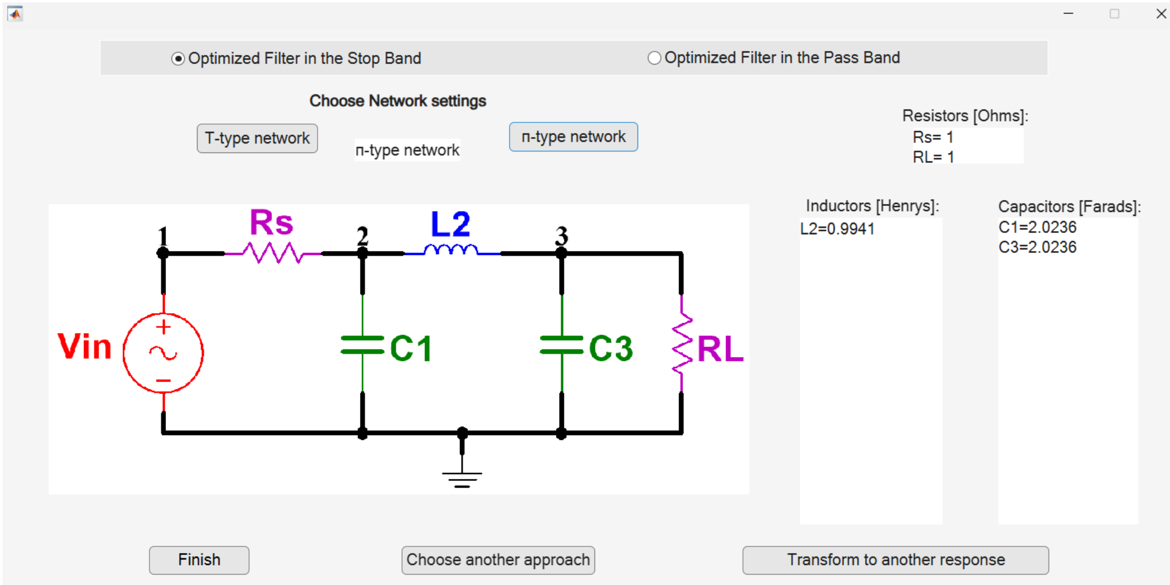
The way of printing the transfer function in the most current versions of MATLAB numerically drags internal structures of the display.

The transfer function obtained is the one registered in Table III of the article in the section Pascal Approximation Method with Optimization in the Stopband, Order 3, LPTF.

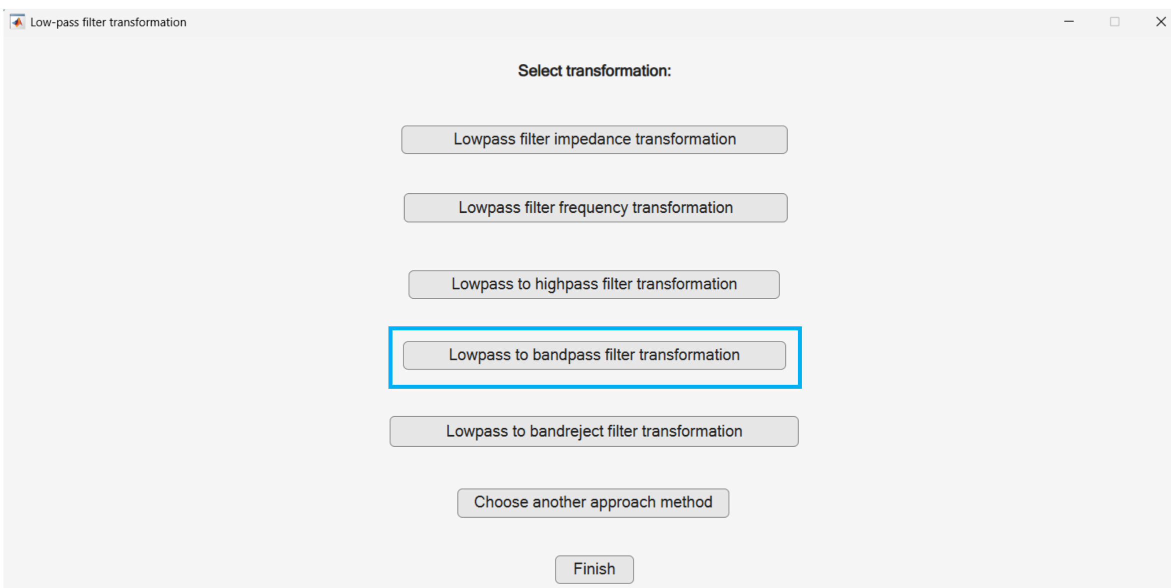
Continuing, the magnitude frequency response is determined.



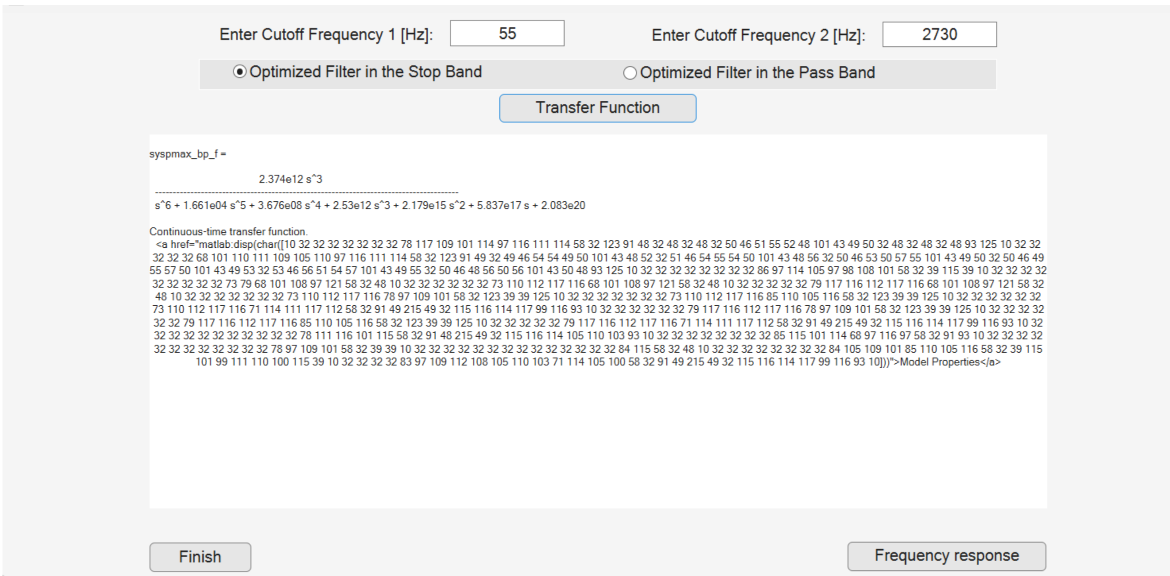
Then the normalized passive electrical network in π configuration.



The transformation from low-pass filter to band-pass filter is chosen.

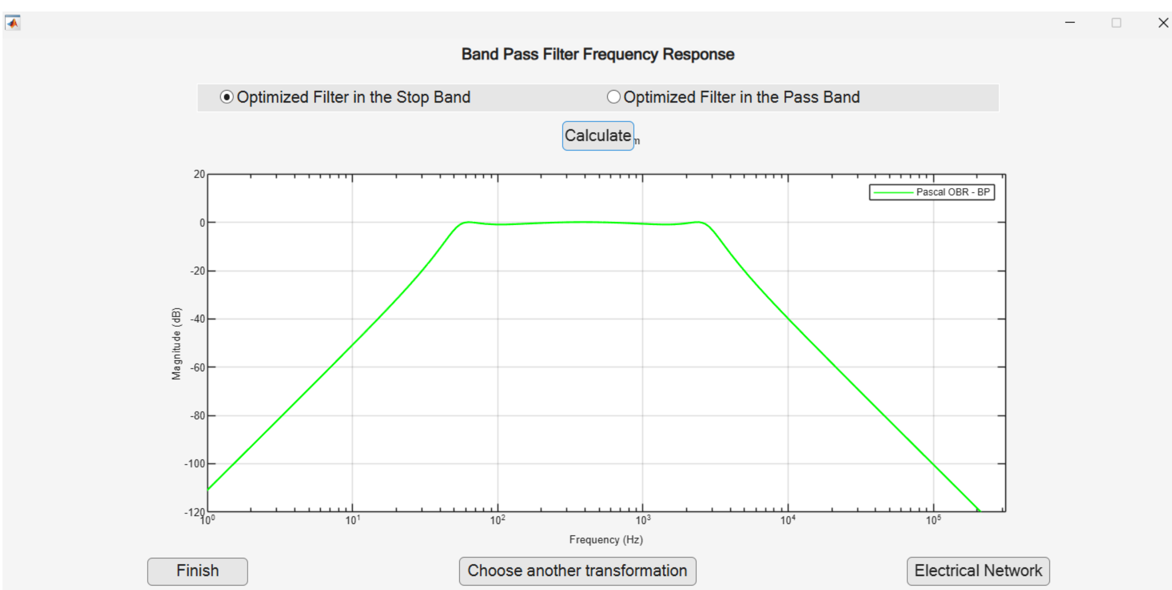


The cutoff frequency specifications are entered where it is intended to obtain a result of 50 Hz and 3000 Hz of cutoff frequencies in the frequency response, to obtain these results a tuning is carried out by trial and error in the entry of cutoff frequencies for the transformation. It is found that by entering the frequencies of 55 Hz and 2730 Hz, the cutoff frequencies measured in the frequency response are very close to 50 Hz and 3000 Hz.

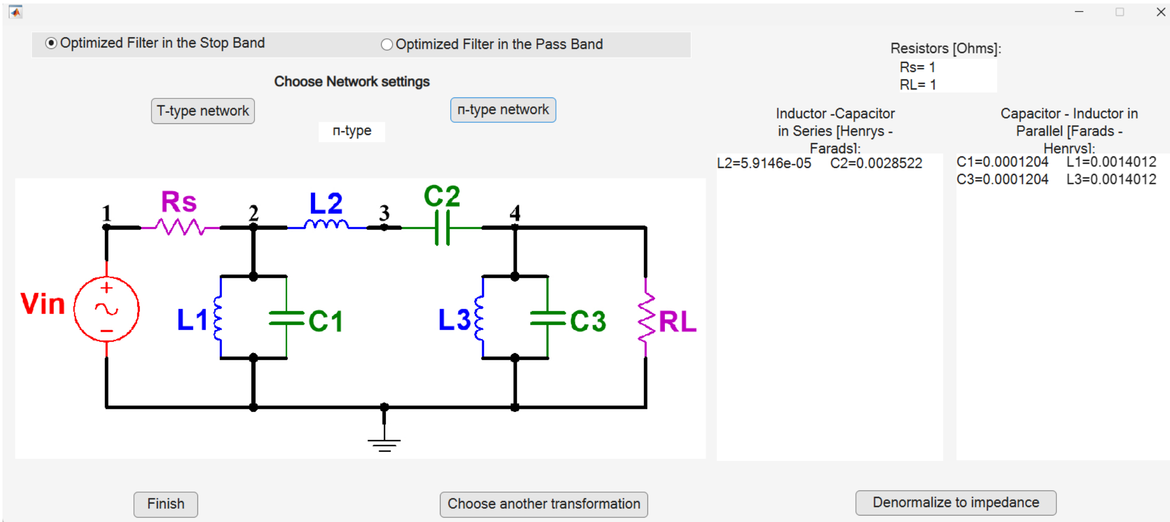


The transfer function registered in Table III of the article in the section Pascal Approximation Method with Optimization in the Stopband, Order 3, BPTF is obtained.

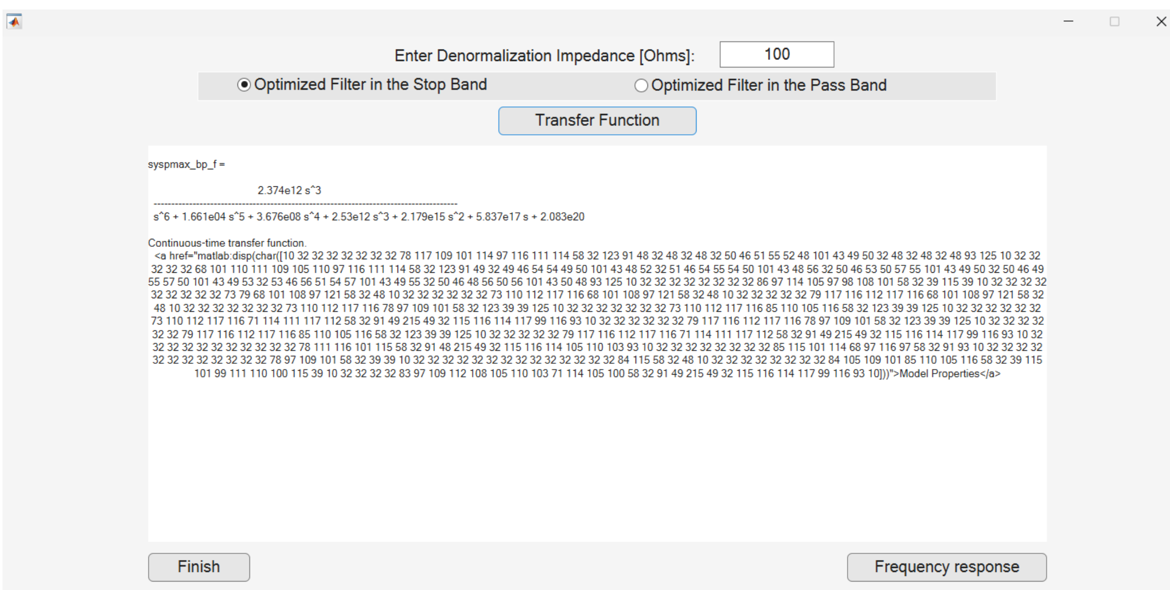
Subsequently the frequency response is generated.



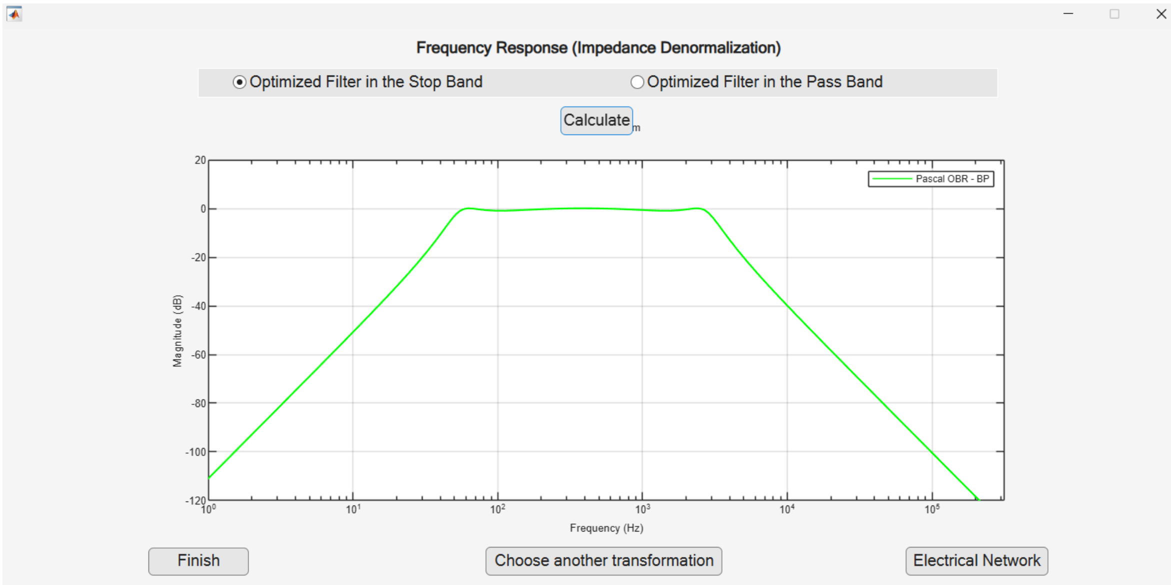
The transformed electrical network is generated.



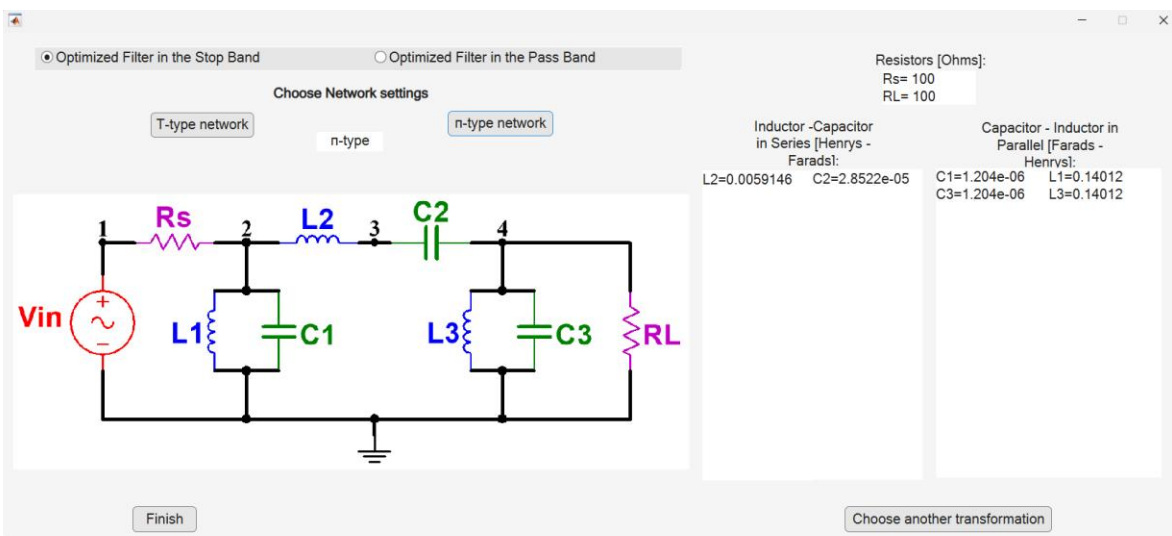
Impedance denormalization is applied with a value of 100Ω.



The transformed frequency response is visualized again.



Finally, the transformed and impedance denormalized electrical network is generated.



The values obtained and the drawing of the electrical network are those shown in Figure 7 of the article, to which the transformation to active elements is externally performed and subsequently it is physically implemented.